Fourier Expansion of $g(w)$

$$g(w) = \sum_{n=-\infty}^{\infty} c_n e^{inw}$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(w) e^{-inw} \, dw$$

Consider

$$c_r = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(w) e^{-irw} \, dw$$

$$\int_{-\pi}^{\pi} g(w) e^{-irw} \, dw = \int_{-\pi}^{\pi} \left(\sum_{n=-\infty}^{\infty} c_n e^{inw} \right) e^{-irw} \, dw$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\pi}^{\pi} c_n e^{i(n-r)w} \, dw$$

(Tonelli's theorem)

$$= \sum_{n=-\infty}^{\infty} c_n \int_{-\pi}^{\pi} e^{i(n-r)w} \, dw \qquad (*)$$

Consider

$$\int_{-\pi}^{\pi} e^{i(n-r)w} \, dw = \int_{-\pi}^{\pi} \cos[(n-r)w] \, dw + i \int_{-\pi}^{\pi} \sin[(n-r)w] \, dw$$

$$\int_{-\pi}^{\pi} e^{i(n-r)w} \, dw = 0$$ except when $n = r$.

when $r = n$

$$\int_{-\pi}^{\pi} dw = 2\pi$$

$\therefore$ In $(*)$ $\displaystyle\int_{-\pi}^{\pi} g(w) e^{-irw} \, dw = 2\pi c_r$

$$\Rightarrow \quad c_r = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(w) e^{-irw} \, dw$$